JTurbo Hydrogen and LNG Liquefaction Technology

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC.
All Rights Reserved.
AV20230057

JTurbo Engineering and Technology

- JTurbo Engineering and Technology

- JTurbo Engineering & Technology is company that specializes in providing process technology and consulting services to the gas processing industries including Hydrogen & LNG liquefaction, LNG regasification, and NGL & other gas processing facilities.
- JTurbo professionals have over 45 years experience in the oil & gas processing, petroleum refining and petrochemicals industries, including Hydrogen and LNG liquefaction technology development
- JTurbo, in collaboration with a Top-tier engineering firm, is currently working towards several 15 & 30-150 tons per day (tpd) hydrogen liquefaction projects for key players in the growing hydrogen economy
-

- JTurbo Liquefaction Technology Patent Applications
- Twin expander N2 precooling and H2 / H2+ Ne, / He refrigeration cycle hydrogen liquefaction
- LNG / LN_2 precooled and twin expander H2 / H+-Ne / He refrigeration cycle hydrogen liquefaction
- Twin Column, High Pressure Absorption Process (TCHAP) C5+ heavies removal
- Turbo-expander NGL recovery processes
- Integrated NGL recovery and LNG liquefaction

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

Advantages of LH_2

- LH_2 has been proven to be the only practical and economical solution for mobility applications after several years of trials.
- LH_2 offers elegant storage/buffer solution for dynamic power supply from renewables for meeting continuous supply of hydrogen for Industrial Applications.
- LH_2 transport offers 7:1 carrier advantage over compressed gas transport.
- LH_2 handling technology is well matured and several products are available from multiple vendors, especially for pumping into high pressure gas for end users.

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

JTurbo Hydrogen Liquefaction Patent (US 11,391,511 B1)

- JTurbo's patent includes fully optimized Twin-expander Nitrogen & Hydrogen cycle.
- Single Train capability: 15, 30 to 150 TPD
- Efficiency: High efficiency 5.2 kW/kg LH_2 compared to current industry practice of > 11 kWh/kg LH_2
- Innovation: Liquefaction plant can be integrated with Air Separation Unit (ASU) or LNG Regasification Terminal. ~30% savings in OPEX and CAPEX.
- Reference: Plug Power, key player in green hydrogen economy, Licensed JTurbo Hydrogen Liquefaction Technology for deployment to Customers
- Application Sectors Attracting due to low specific energy consumption (SEC): Data Center Development, Ammonia Production, Methanol Production, and Steel Industry

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

JTurbo Technology Details

- JTurbo provides lowest specific energy consumption (SEC) [kWh per kg of liquid hydrogen]
  - Significantly lower SEC (50% advantage) compared to current liquefaction technologies available in the industry.

- Built upon established thermodynamic cycles and equipment technologies
  - H2 Precooling and H2 Liquefaction stages use the proven Reverse-Brayton thermodynamic refrigeration cycles
  - Twin expanders, plate-fin heat exchangers, and refrigerant compressors in each stage support high efficiency, proven equipment design available from established, reliable vendors
  - Nitrogen and Hydrogen are established refrigerants for the two stages thermodynamic cooling cycles.
    - Other refrigerants such as Hydrogen / Helium / H2-Neon-mix can be used without losing efficiency when needed.

JTurbo H_2 Pre-Cooling
N_2 Refrigerant
Reverse-Brayton Cycle
Twin Expander

→

H_2 Purification
Cryogenic Adsorption
(as needed)

→

JTurbo H_2 Liquefaction
H_2 Refrigerant
Reverse-Brayton Cycle
Twin Expander

→

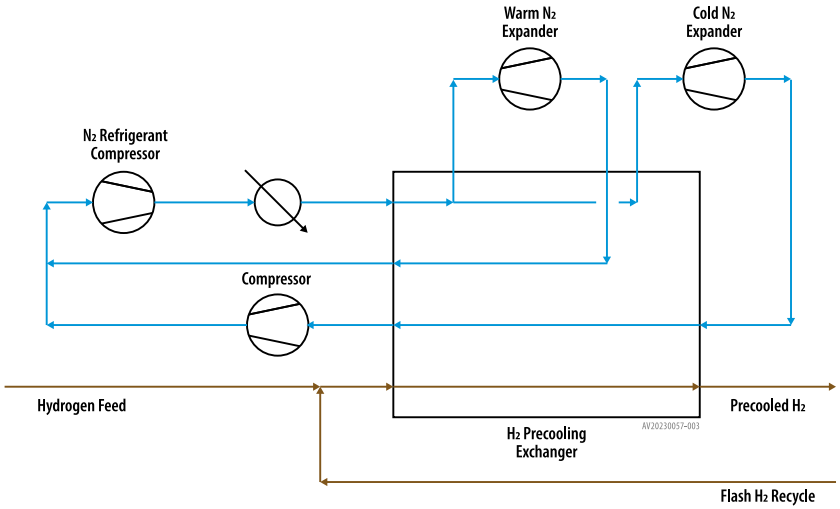
Liquid H_2 Storage
Vacuum Insulated
Sphere or Bullets

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

JTurbo Technology Details

Highly Efficient Design, Proven Elements

- First stage uses a closed-loop nitrogen refrigeration cycle
  - Proven Reverse-Brayton thermodynamic refrigeration cycle with twin turbo-expanders for high efficiency
  - Demonstrated plate-fin heat exchanger technology for efficient cooling
  - Well-established refrigerant compressor technology
  - Water-cooling for final heat rejection keeps efficiency high
    - Actual efficiency depends on heat sink for ambient cooling
  - Alternate utilization of free cold energy from LNG / LN2 vaporization in LNG regasification terminal and Air Separation Units reduce CAPEX and OPEX significantly by 30%.

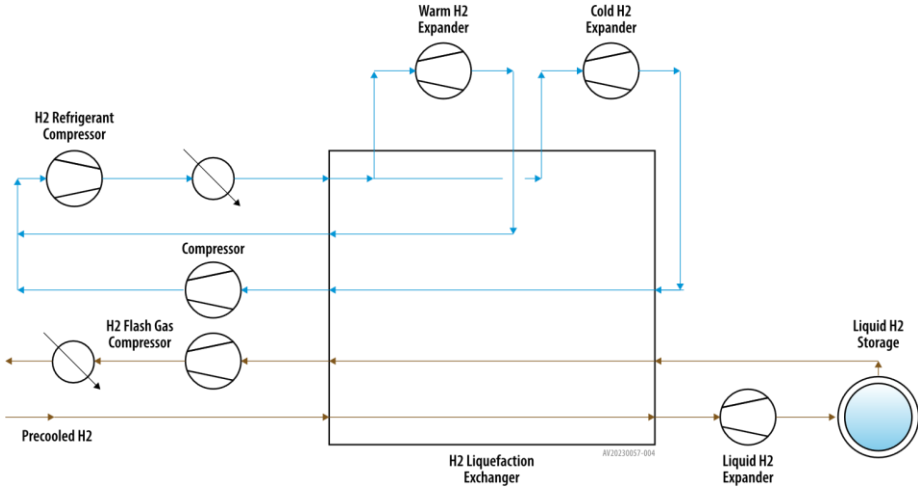


© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

Patented JTurbo's Leading-Edge Hydrogen Liquefaction Technology

Highly Efficient Design, Proven Elements

- Second stage uses a closed-loop hydrogen refrigeration cycle
  - Proven Reverse-Brayton thermodynamic refrigeration cycle with multiple turbo-expanders for high efficiency
  - Alternate He or H2-Ne mixed refrigerants
  - Demonstrated plate-fin heat exchanger technology for efficient cooling
  - Well-established refrigerant compressor technology
  - Water-cooling for final heat rejection keeps efficiency high
    - Actual efficiency depends on heat sink for ambient cooling



© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

JTurbo Technology Details

Medium and Large-Scale Hydrogen Liquefaction Facilities

- Comparisons of core-process specific energy consumption (SEC)
  - JTurbo: As low as 5.2 kWh per kg liquid-H2 SEC (SEC will slightly vary based on H2 supply pressure and cooling water limitations) Vs Others: 11 + kWh per kg liquid-H2 (existing plants and those under development)
- Economically viable medium and large-scale hydrogen liquefaction processes
  - Optimized total cost of ownership (TCO) due to low CAPEX and OPEX
- Plant Capacity
  - 15, 30, 50 , 100 and 150 ton-per-day (TPD) designs are available, Larger capacity, need multiple trains
- Continuous improvement
  - Engagement with OEMs in the development of dynamic gas bearing and high-speed turbo expander-compressors
  - Potential to improve the isentropic efficiency by 5 - 10% and reduce SEC below 5.0 kW/kg

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

Hydrogen Liquefaction
Current Technology Comparison

Comparing SEC for existing plants and JTurbo Liquefaction Technology using a consistent design basis

Current Hydrogen Liquefaction Technology and Proposed New Developments					
Technology	**Status**	**Pre-cooling Cycle**	**Liquefaction & Sub-cooling Cycle**	**Specific Energy Consumption (SEC) kWh/kg**	**Exergy Efficiency (%)**
Linde–Ingolstadt	Operating	LN2	H2 - Claude	13.60	28.82
Linde–Leuna	Operating	LN2	H2 - Claude	11.90	32.94
Air Products	6 × operating	LN2	H - Claude	12 - 15	33 - 26
Praxair	4 × operating	LN2	H2 - Claude	12.5 - 15	31 - 26
Air Liquide	5 × operating	LN2	H2 - Claude	12 - 15	33 - 26
JTURBO	Patent No. US 11,391,511 B1	N2 - Reverse Brayton	He - Reverse Brayton	5.35	73.22
		N2 - Reverse Brayton	H2 - Reverse Brayton	5.29	74.10
		N2 - Reverse Brayton	H2+Ne - Reverse Brayton	5.20	75.46
		LNG	H2+Ne - Reverse Brayton	3.61	108.74

Notes:

1. H_2 Plant Feed Gas & Liquefaction Pressure @ 1160 psi (80 bar) and H_2 Storage @ 14.5 psi (1.0 bar)

2. Specific Energy Consumption (SEC) does not include a constant value of 527 kJ/kg, Ortho-Para H_2 Conversion (OPC) heat duty for consistency with other Technologies that has not been included in the published SEC.

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

JTurbo Hydrogen Liquefaction Technology

JTurbo Hydrogen Liquefaction Technology Summary

Hydrogen Liquefaction & Refrigeration Summary (Average Ambient Temperature 25 °C)					
Pre-Cooling Cycle	Twin-Exp N_2 [1] (1+1) Cycle	Twin-Exp N_2 [2] (1+1) Cycle	Twin-Exp N_2 [3] (1+1) Cycle	LNG [4]	LN_2 [5]
Liquefaction & Subcooling Refrigeration Cycle	Twin-Exp H_2 (1+1) Cycle	Twin-Exp H_2 (2+1) Cycle	Twin-Exp H_2 (3+2) Cycle	Twin-Exp H_2 (3+2) Cycle	Twin-Exp H_2 (3+2) Cycle
Hydrogen Feed Gas (Nm3/h)	23,168	46,331	69,491	69,491	69,491
Liquid Hydrogen Loading (ton/day)	50	100	150	150	150
Specific Energy Consumption (kWh/kg)	6.9	6.0	5.7	4.5	3.6
Exergy Efficiency (%)	57.1	65.3	68.5	87.0	108.0
Total Power (kWh)	14,304	25,011	35,799	28,402	22,722
Power Balance (%)	1.20	1.05	1.00	0.79	0.63
Total CAPEX (US $ / kg LH2 / Day	1,300	1,000	700	600	550

Notes

[1] Twin-Exp JTurbo N_2 / H_2 (1+1) Exp H_2 Cycle: Twin-Exp JTurbo Nitogen Precooling and H_2 (1+1) Exp Cycle H_2 Liquefaction

[2] Twin-Exp JTurbo N_2 / H_2 (2+1) Exp H_2 Cycle: Twin-Exp JTurbo Nitogen Precooling and H_2 (2+1) Exp Cycle H_2 Liquefaction

[3] Twin-Exp JTurbo N_2 / H_2 (3+2) Exp H_2 Cycle: Twin-Exp JTurbo Nitogen Precooling and H_2 (3+2) Exp Cycle H_2 Liquefaction

[4] LNG / H_2 (3+2) Exp H_2 Cycle: Twin-Exp JTurbo LNG Precooling and H_2 (3+2) Exp Cycle H_2 Liquefaction

[5] LN_2 / H_2 (2+1) Exp H_2 Cycle: Twin-Exp JTurbo LN2 Precooling and H_2 (2+1) Exp Cycle H_2 Liquefaction

* H_2 Feed Gas from H_2 Storage & H_2 Liquefaction @ 580 psig (40 bar) and H_2 Storage @ - 418.4 °F (-250 °C) and 29 psig (2 bar)

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

JTurbo LH_2 Storage Solution for Dta Center, Green Ammonia, Steel, Refinery, Methanol…..Plants

LH2 Storage Solution for Data Center,Green Ammonia, Steel, Refinery Methanol, … Plants

- JTurbo's cost effective solution is based on the integration of the Data Center, Ammonia,Steel, Refinery , Methanol, …. Plant and the LH_2 Plant
- During daytime, H_2 from Electrolyzer are fed directly to the Industrial Plants Ammonia Plant using Renewable Energy. Excess H_2 from Electrolyzer is Liquefied and stored as LH_2 in a Cryogenic Tank for nighttime operation
- During nighttime, LH_2 from the Tank is vaporized and fed to Industrial plants while the LH_2 plant is on recycle mode, producing Liquid Air/ Nitrogen & stored in a Tank for LH_2 Precooling cycle (daytime) from free cold energy exchange
- Cold H_2 gas from LH_2 vaporizer will be used as cooling media in the Chiller for Data Centers, Fuel Cells and other Industrial Application
- Above LH_2 Plant integration, whereby free cold energy from LH_2 vaporization can be effectively utilized for Liquefying Air / N_2 for significant LH_2 Plant CAPEX and OPEX savings by 30%

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

Hydrogen Liquefaction Technology
LH_2 Carriers

Using Liquid Hydrogen for >1,000 Mile long haul Trucks →

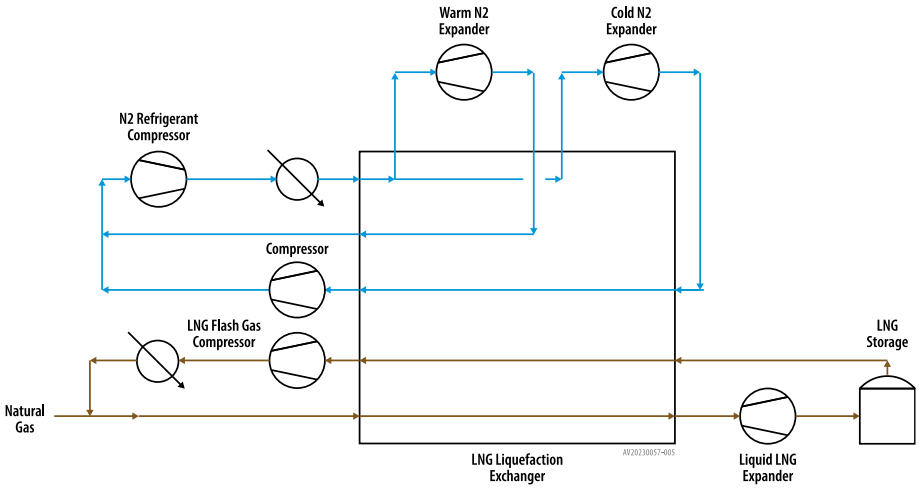
Liquid Hydrogen carriers (4 x 40,000 m^3)→

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

Patented JTurbo's Leading-Edge LNG Liquefaction Technology

Highly Efficient Design, Proven Elements

- Proven Reverse-Brayton thermodynamic refrigeration cycle with twin turbo-expanders for high efficiency
- Demonstrated plate-fin heat exchanger technology for efficient cooling
- Well-established refrigerant compressor technology
  - Water-cooling for final heat rejection keeps efficiency high
    - Actual efficiency depends on heat sink for ambient cooling



© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

LNG Liquefaction + Refrigeration Summary
1.0 MTA LNG Liquefaction System

JTurbo's Patented SMR & Twin Expander Nitrogen cycle LNG Liquefaction Technology takes the lead, offering an Industry-low Refrigerant Power 250 - 300 kWh/ ton LNG and a highly competitive CAPEX of $ 350 per ton LNG , compared to the Industry's prevailing figures of 300- 500 kWh/ ton LNG and $ 800 + per ton LNG.

LNG Liquefaction & Refrigeration Summary (Average Ambient Temperature 25 ^{o}FC)			
JTURBO	Description	SMR	TEXP-N_2 Cycle
Patent No. US 12,,181,214 B2	Treated Feed Gas (MMSCFD)	134.5	128.6
	LNG Loading (TPD)	3,062	2,909
	LNG Loading (MTA)	1.118	1.062
	Specific Refrigerant Power (kW/Ton)	278	300
	Total Refrigerant Power (kW)	35,515	36,334
	Power Balance (%)	1.00	1.08

Other LNG Liquefaction Technology	
APCI SMR Specific Refrigerant Power (kW/Ton)	300
Other Gaseous Nitrogen Expander Cycle Specific Refrigerant Power (kW/Ton)	500

Legend
SMR : Single Mixed Refrigerant Cycle
TEXP-N_2 : Twin Expander Nitrogen Cycle

JTurbo Engineering & Technology, LLC

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

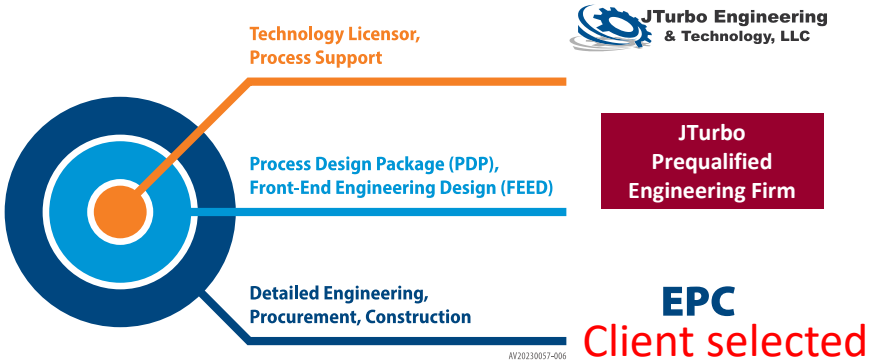
JTurbo's Execution Approach

JTURBO: PRODUCTIZING MOST ENERGY EFFICIENT HYDROGEN & LNG LIQUEFACTION TECHNOLOGY

JTurbo licenses and delivers the liquefaction technology including the material balance, process flowsheet, and critical specifications.

JTurbo engages Top-tier EPC firm to complete and deliver an appropriately detailed Process Design Package (PDP) or Front-End Engineering Design Package (FEED) based on project needs.

This execution approach brings state-of-the-art technology together with quality engineering to deliver cost-effective, efficient, and reliable hydrogen liquefaction technologies.



© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057

Thank you

For additional information, please contact

Jacob Thomas
JTurbo Engineering and Technology LLC
4607 Round Rock Court
Sugar Land, Texas 77479, U.S.A

O: + 1. 713.838.2375
M: + 1. 281.851.9598

E: JThomas@JTurboengineering.com
E: Inquiries@JTurboengineering.com
www.JTurboengineering.com

© 2023 JTurbo Engineering & Technology, LLC. All Rights Reserved.
AV20230057